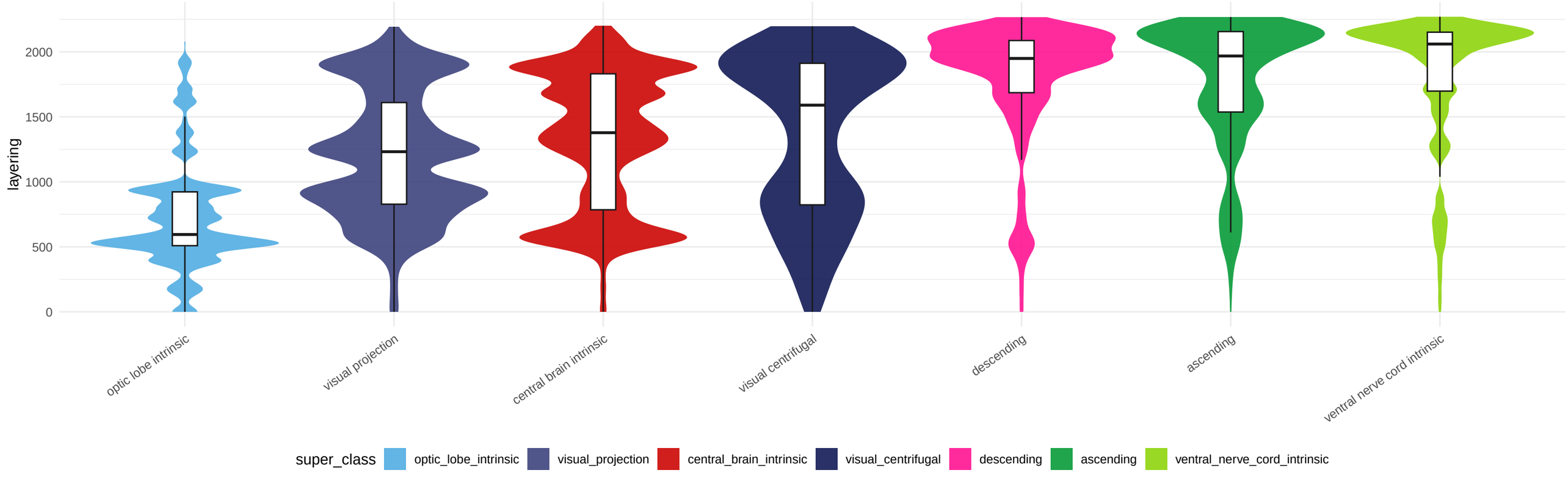
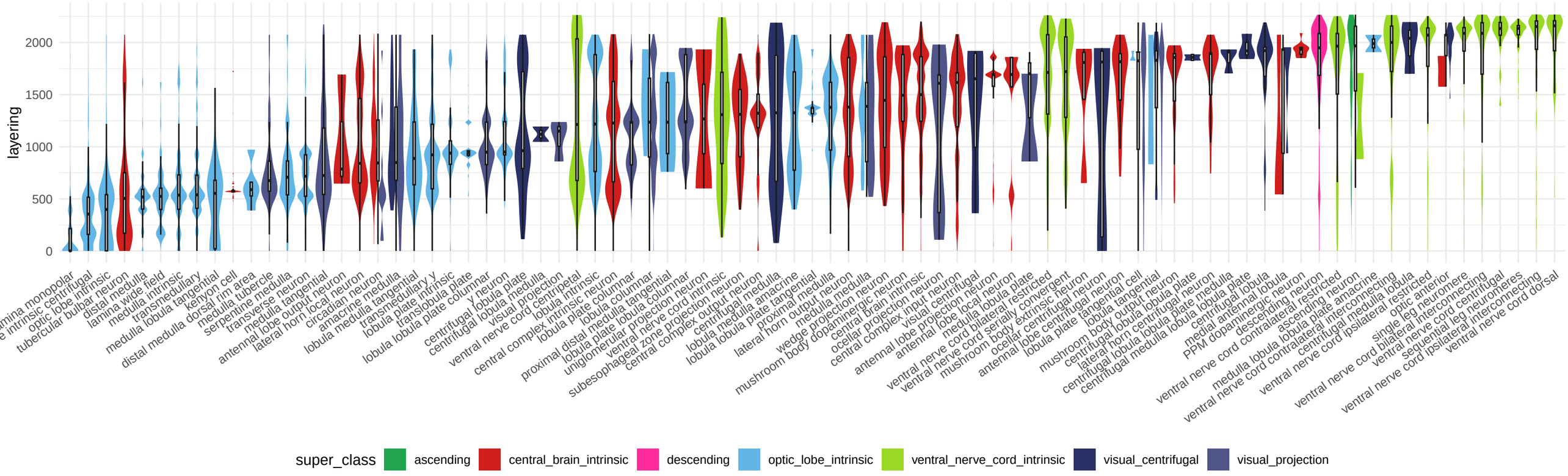

layering
2000
1500
1000
500
0
optic lobe intrinsic
visual projection
central brain intrinsic
visual centrifugal
descending
ascending
ventral nerve cord intrinsic
super_class
optic_lobe_intrinsic
visual_projection
central_brain_intrinsic
visual_centrifugal
descending
ascending
ventral_nerve_cord_intrinsic


layering
2000
1500
1000
500
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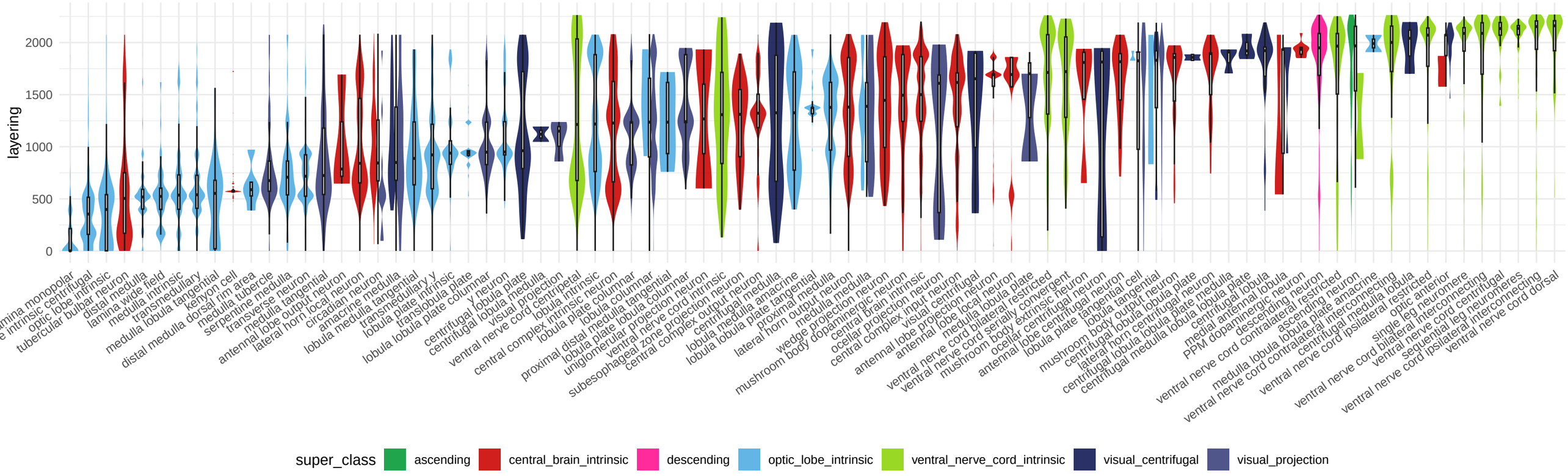
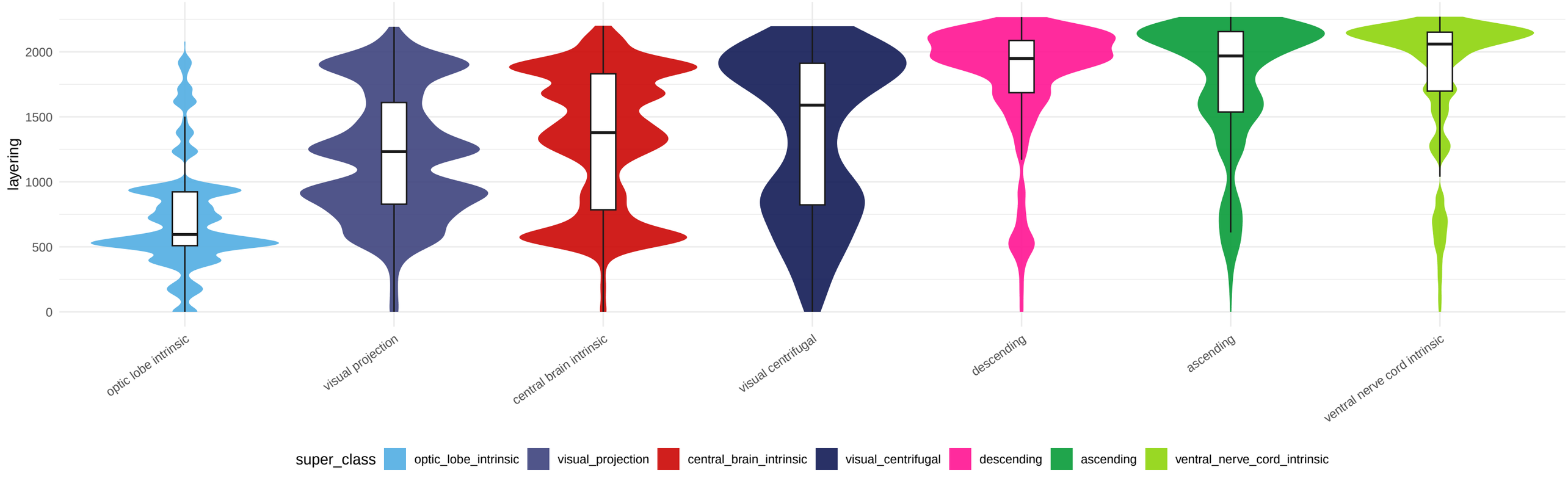
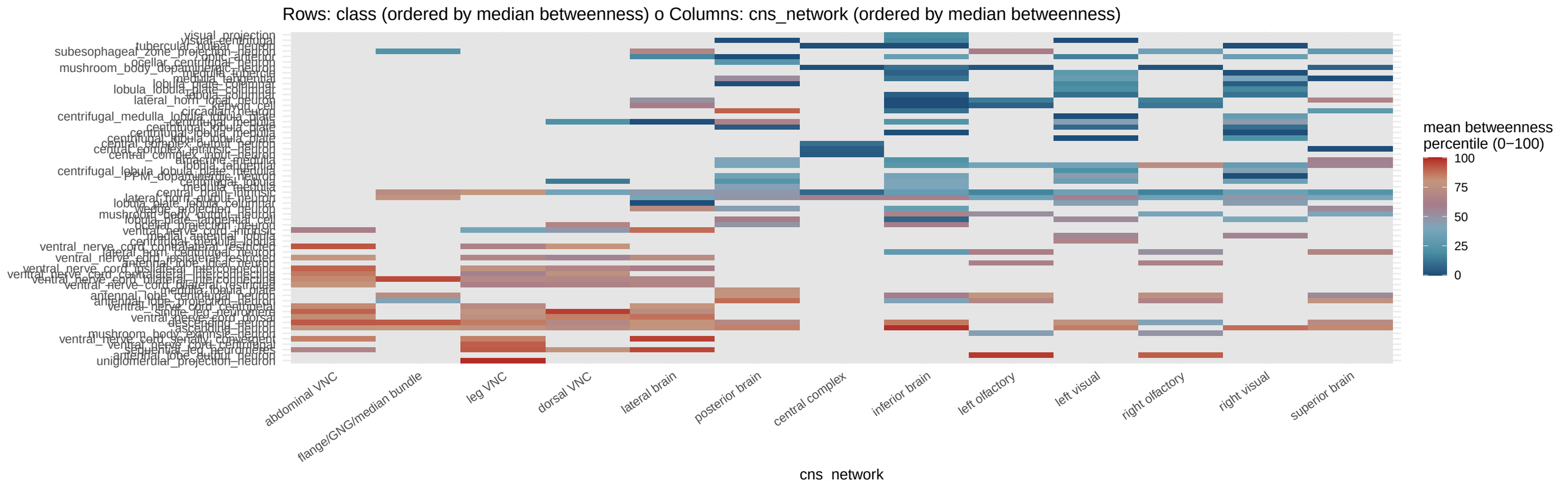
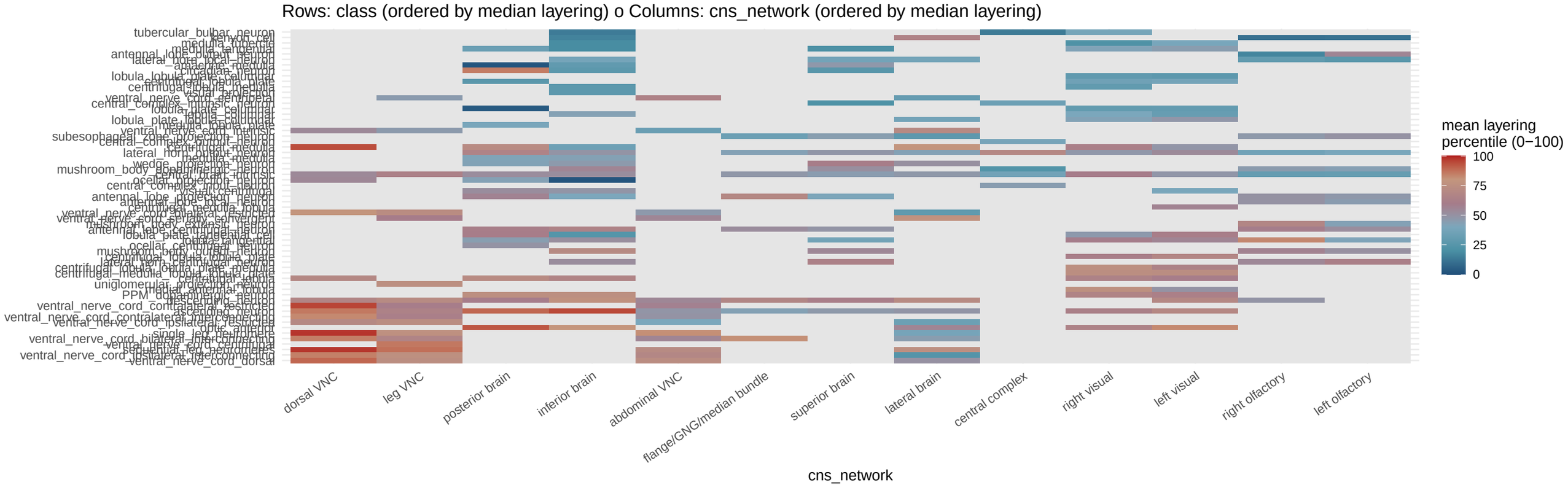
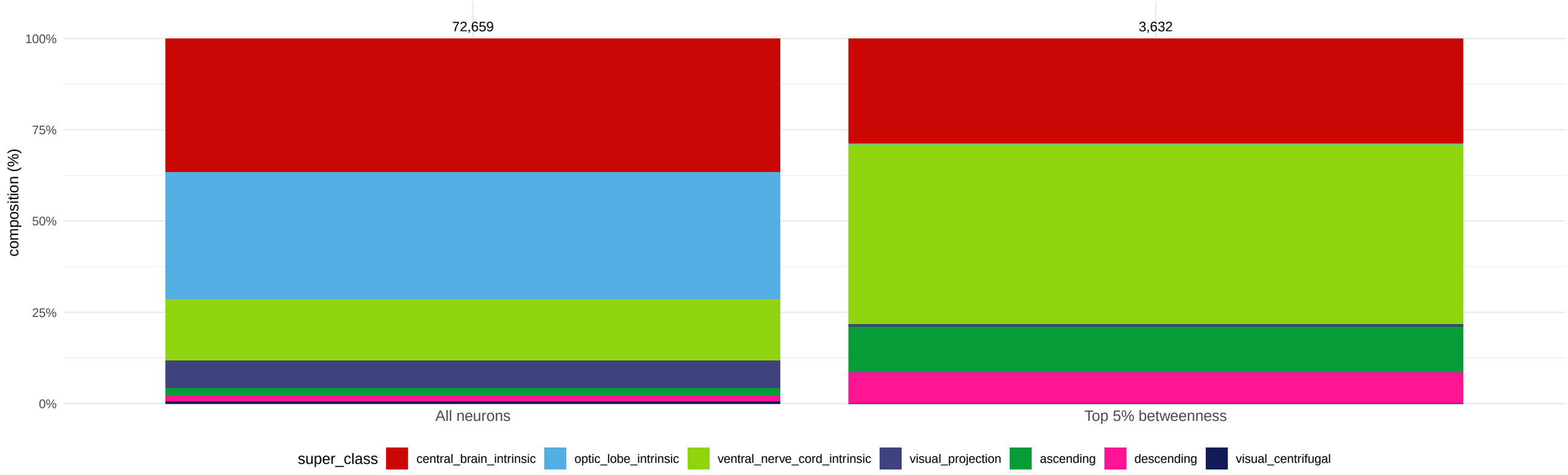
amina monopolar
e intrinsic centrifugal
optic lobe intrinsic
tubercular bulbar neuron
distal medulla
lamina wide field
medulla intrinsic
transmedullary
medulla lobula tangential
kenyon cell
distal medulla dorsal rim area
medulla tubercle
serpentine medulla
transverse neuron
medulla tangential
antennal lobe output neuron
lateral horn local neuron
circadian neuron
amacrine neuron
lobula medulla tangential
transmedullary y
lobula plate intrinsic
translobula plate
lobula lobula plate columnar
y neuron
centrifugal lobula plate
centrifugal lobula medulla
visual projection
ventral nerve cord centripetal
central complex intrinsic
lobula intrinsic neuron
lobula plate columnar
lobula columnar
proximal distal medulla tangential
lobula plate lobula columnar
uniglomerular projection neuron
ventral nerve cord intrinsic
subesophageal zone projection neuron
central complex output neuron
centrifugal medulla
lobula medulla amacrine
lobula lobula plate tangential
lateral horn output neuron
proximal medulla
wedge projection neuron
mushroom body dopaminergic neuron
medulla medulla
central brain intrinsic
ocellar projection neuron
central complex input neuron
visual centrifugal
antennal lobe projection neuron
medulla lobula plate
antennal lobe local neuron
ventral nerve cord bilateral restricted
ventral nerve cord serially convergent
mushroom body extrinsic neuron
ocellar centrifugal neuron
antennal lobe centrifugal neuron
lobula plate tangential cell
lobula tangential
mushroom body output neuron
centrifugal lobula lobula plate
lateral horn centrifugal neuron
centrifugal lobula plate medulla
centrifugal medulla lobula lobula
medial antennal lobula
PPM dopaminergic neuron
descending neuron
ventral nerve cord contralateral restricted
ascending neuron
medulla lobula lobula plate amacrine
ventral nerve cord contralateral interconnecting
centrifugal interconnecting
ventral nerve cord ipsilateral restricted
optic anterior
single leg neuromere
ventral nerve cord bilateral interconnecting
ventral nerve cord centrifugal
sequential leg neuromeres
ventral nerve cord ipsilateral interconnecting
ventral nerve cord dorsal
super_class
ascending
central_brain_intrinsic
descending
optic_lobe_intrinsic
ventral_nerve_cord_intrinsic
visual_centrifugal
visual_projection

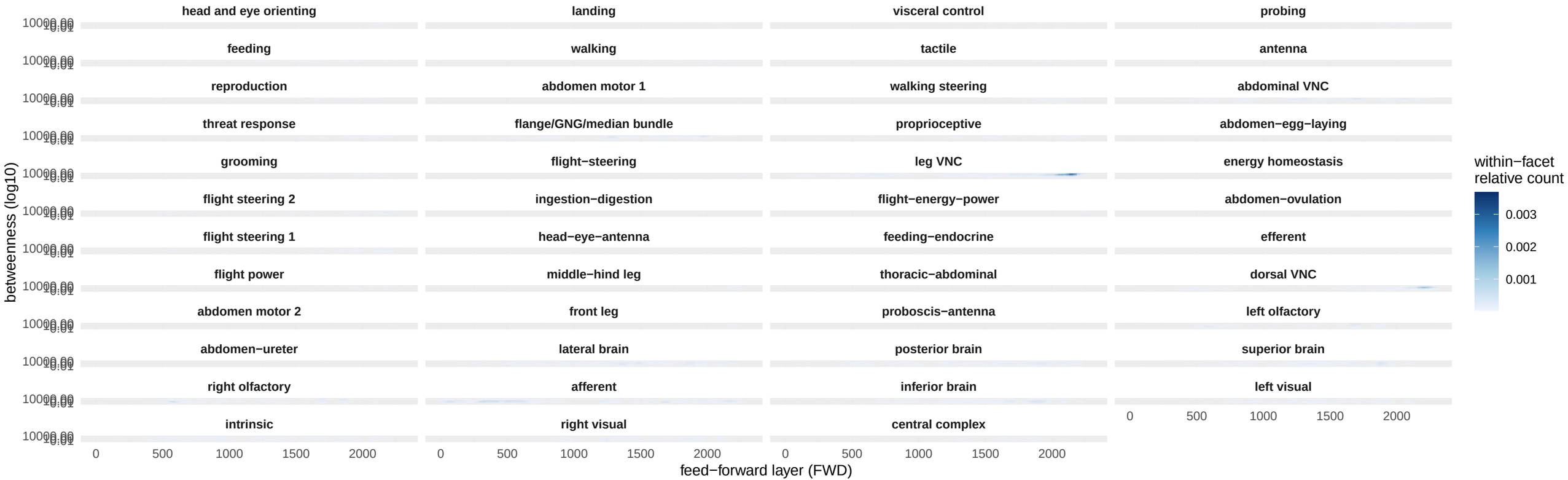

layering
2000
1500
1000
500
0
amina monopolar
e intrinsic centrifugal
optic lobe intrinsic
tubercular bulbar neuron
distal medulla
lamina wide field
medulla intrinsic
transmedullary
medulla lobula tangential
kenyon cell
distal medulla dorsal rim area
medulla tubercle
serpentine medulla
transverse neuron
medulla tangential
antennal lobe output neuron
lateral horn local neuron
circadian neuron
amacrine neuron
lobula medulla tangential
transmedullary y
lobula plate intrinsic
translobula plate
lobula lobula plate columnar
y neuron
centrifugal lobula plate
centrifugal lobula medulla
visual projection
ventral nerve cord centripetal
central complex intrinsic
lobula intrinsic neuron
lobula plate columnar
lobula columnar
proximal distal medulla tangential
lobula plate lobula columnar
uniglomerular projection neuron
ventral nerve cord intrinsic
subesophageal zone projection neuron
central complex output neuron
centrifugal medulla
lobula medulla amacrine
lobula lobula plate tangential
lateral horn output neuron
proximal medulla
wedge projection neuron
mushroom body dopaminergic neuron
medulla medulla
central brain intrinsic
ocellar projection neuron
central complex input neuron
visual centrifugal
antennal lobe projection neuron
medulla lobula plate
antennal lobe local neuron
ventral nerve cord bilateral restricted
ventral nerve cord serially convergent
mushroom body extrinsic neuron
ocellar centrifugal neuron
antennal lobe centrifugal neuron
lobula plate tangential cell
lobula tangential
mushroom body output neuron
centrifugal lobula lobula plate
lateral horn centrifugal neuron
centrifugal lobula plate medulla
centrifugal medulla lobula lobula
medial antennal lobula
PPM dopaminergic neuron
descending neuron
ventral nerve cord contralateral restricted
ascending neuron
medulla lobula lobula plate amacrine
ventral nerve cord contralateral interconnecting
centrifugal interconnecting
ventral nerve cord ipsilateral restricted
optic anterior
single leg neuromere
ventral nerve cord bilateral interconnecting
ventral nerve cord centrifugal
sequential leg neuromeres
ventral nerve cord ipsilateral interconnecting
ventral nerve cord dorsal
super_class
ascending
central_brain_intrinsic
descending
optic_lobe_intrinsic
ventral_nerve_cord_intrinsic
visual_centrifugal
visual_projection


layering
2000
1500
1000
500
0
optic lobe intrinsic
visual projection
central brain intrinsic
visual centrifugal
descending
ascending
ventral nerve cord intrinsic
super_class
optic_lobe_intrinsic
visual_projection
central_brain_intrinsic
visual_centrifugal
descending
ascending
ventral_nerve_cord_intrinsic


Rows: class (ordered by median betweenness) o Columns: cns_network (ordered by median betweenness)
visual_projection
visual_centrifugal
tubercular_pontar_neuron
subesophageal_zone_projection_neuron
ocellar_centrifugal_neuron
mushroom_body_dopaminergic_neuron
medulla_tangential
lobula_plate_columnar
lobula_lobula_plate_columnar
lobula_columnar
lateral_horn_local_neuron
Kenyon_cell
circadian_neuron
centrifugal_medulla_lobula_plate_tangential
centrifugal_medulla
centrifugal_lobula_plate
centrifugal_lobula_medulla
centrifugal_lobula_lobula_plate
central_complex_output_neuron
central_complex_intrinsic_neuron
central_complex_input_neuron
ellipsoid_neuron
centrifugal_lobula_lobula_plate_medulla
PPM_dopaminergic_neuron
centrifugal_lobula
medulla_intrinsic
central_brain_intrinsic
lateral_horn_output_neuron
lobula_plate_lobula_columnar
wedge_projection_neuron
mushroom_body_output_neuron
lobula_plate_tangential_cell
pushing_projection_neuron
ventral_nerve_cord_intrinsic
medial_lateral_lobula
centrifugal_medulla_lobula
ventral_nerve_cord_contralateral_neuron
ventral_nerve_cord_lateral_neuron
ventral_nerve_cord_bilateral_restricted
antennal_lobe_local_neuron
ventral_nerve_cord_bilateral_interconnecting
ventral_nerve_cord_contralateral_interconnecting
ventral_nerve_cord_bilateral_restricted
medulla_lobula_plate
antennal_lobe_central_neuron
antennal_lobe_projection_neuron
ventral_nerve_cord_commissural
single_leg_neuromere
ventral_nerve_cord_dorsal
descending_neuron
ascending_neuron
mushroom_body_extrinsic_neuron
ventral_nerve_cord_sensory_convergent
ventral_nerve_cord_centrifugal
sensorial_leg_neuromere
antennal_lobe_output_neuron
uniglomerular_projection_neuron
abdominal VNC
flange/GNG/median bundle
leg VNC
dorsal VNC
lateral brain
posterior brain
central complex
inferior brain
left olfactory
left visual
right olfactory
right visual
superior brain
cns_network
mean betweenness percentile (0−100)
100
75
50
25
0


Rows: class (ordered by median layering) o Columns: cns_network (ordered by median layering)
tubercular_bulbar_neuron
kenyon_cell
medulla_tubercle
medulla_tangential
antennal_lobe_output_neuron
lateral_horn_local_neuron
amacrine_medulla
centrifugal_neuron
lobula_lobula_plate_columnar
centrifugal_lobula_plate
centrifugal_lobula_medulla
visual_projection
ventral_nerve_cord_centrifugal
central_complex_intrinsic_neuron
lobula_plate_columnar
lobula_plate_lobula_columnar
lobula_medulla_lobula_plate
ventral_medulla_lobula_plate
ventral_nerve_cord_intrinsic
subesophageal_zone_projection_neuron
central_complex_output_neuron
centrifugal_medulla
lateral_horn_output_neuron
medulla_medulla
wedge_projection_neuron
mushroom_body_cholinergic_neuron
central_brain_intrinsic
ocellar_projection_neuron
central_complex_output_neuron
visual_centrifugal
antennal_lobe_projection_neuron
antennal_lobe_local_neuron
centrifugal_medulla_lobula
ventral_nerve_cord_sensory_afferent
ventral_nerve_cord_sensory_efferent
mushroom_body_extrinsic_neuron
antennal_lobe_centrifugal_neuron
lobula_plate_tangential_cell
lobula_tangential
ocellar_centrifugal_neuron
mushroom_body_output_neuron
centrifugal_lobula_lobula_plate
lateral_horn_centrifugal_neuron
centrifugal_lobula_lobula_plate_medulla
centrifugal_medulla_lobula_lobula_plate
uniglomerular_projection_neuron
centrifugal_lobula
medial_antennal_lobula
PPM_dopaminergic_neuron
descending_neuron
ventral_nerve_cord_contralateral_restricted
ventral_nerve_cord_contralateral_interconnecting
ascending_neuron
ventral_nerve_cord_ipsilateral_restricted
optic_afferent
single_leg_neuromere
ventral_nerve_cord_bilateral_interconnecting
ventral_nerve_cord_centrifugal
segmental_leg_neuromere
ventral_nerve_cord_ipsilateral_interconnecting
ventral_nerve_cord_dorsal
dorsal VNC
leg VNC
posterior brain
inferior brain
abdominal VNC
flange/GNG/median bundle
superior brain
lateral brain
central complex
right visual
left visual
right olfactory
left olfactory
cns_network
mean layering percentile (0−100)
100
75
50
25
0


72,659
3,632
100%
75%
50%
25%
0%
composition (%)
All neurons
Top 5% betweenness
super_class
central_brain_intrinsic
optic_lobe_intrinsic
ventral_nerve_cord_intrinsic
visual_projection
ascending
descending
visual_centrifugal


head and eye orienting
landing
visceral control
probing
feeding
walking
tactile
antenna
reproduction
abdomen motor 1
walking steering
abdominal VNC
threat response
flange/GNG/median bundle
proprioceptive
abdomen−egg−laying
grooming
flight−steering
leg VNC
energy homeostasis
flight steering 2
ingestion−digestion
flight−energy−power
abdomen−ovulation
flight steering 1
head−eye−antenna
feeding−endocrine
efferent
flight power
middle−hind leg
thoracic−abdominal
dorsal VNC
abdomen motor 2
front leg
proboscis−antenna
left olfactory
abdomen−ureter
lateral brain
posterior brain
superior brain
right olfactory
afferent
inferior brain
left visual
intrinsic
right visual
central complex
0
500
1000
1500
2000
betweenness (log10)
feed−forward layer (FWD)
within−facet
relative count
0.003
0.002
0.001